

Individual Exercise/Assignment One

INSTRUCTIONS

- Total mark in this Exercise/Assignment is 50%, which will account for 5% of the overall grade in this unit.
- Answer **all** questions. Type out your answers (**Times New Roman, font size 12**), but diagrams can be hand drawn. Screenshot hand-drawn diagrams are fine.
- You must submit this document in the **PDF format** along with an **Excel spreadsheet** on Moodle.
- You **must not use generative artificial intelligence (AI)** to generate any materials or content in relation to the assessment task.

QUESTION 1 (DATA APPLICATION): ESTIMATING OKUN'S LAW AND POTENTIAL GDP GROWTH. (15%)

Instruction: You must submit your Excel spreadsheet to show the data values and scatter plot charts as evidence of your working effort.

As explained in Topic 2, Okun's Law states that:

$$\Delta Y/Y = \Delta Y^*/Y^* - \omega \cdot (\Delta U) \text{ ----- Equation (1)}$$

where $\Delta Y/Y$ denotes the real GDP growth rate, $\Delta Y^*/Y^*$ denotes the growth rate of potential real GDP, $-\omega$ denotes Okun's law coefficient, and ΔU denotes the change in unemployment rate. Note that in practice, the Okun's law coefficients vary across countries and tend to differ from the original Okun's law coefficient proposed by Arthur Okun in 1962 given in the standard macroeconomic textbook. This is because the result for Okun's law coefficient is often sensitive to data and may differ due to differences in the labor market structures across countries.

The Okun's Law in Equation (1) is useful for many applications. For example, it can be used to estimate the growth rate of potential real GDP $\Delta Y^*/Y^*$ (which is not directly observable). Since we have data for the change in unemployment rate ΔU and the real GDP growth rate $\Delta Y/Y$, we can estimate the other parameters in Equation (1). Equation (1) is simply the equation of a line ($y = b + mx$, with $y = \Delta Y/Y$, vertical intercept $b = \Delta Y^*/Y^*$, slope $m = -\omega$, and $x = \Delta U$). If we plot ΔU vs. $\Delta Y/Y$, we should be able to estimate the line that best goes through these points. Technically, this is called a "regression line." You are going to use an Excel spreadsheet to estimate this line; Excel calls the line a "trendline." In this question, you will estimate the Okun's line in Equation (1) for Japan.

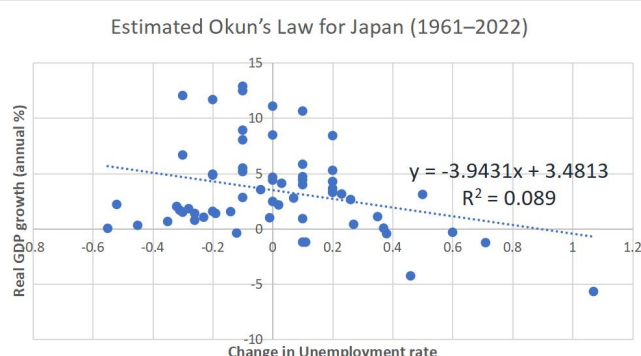
(a) Before you can estimate the Okun's line in Equation (1) for Japan, calculate the change in unemployment rate in Column C of the given Excel spreadsheet titled "Japan data_Stu_Okun Law". Note: The change in unemployment rate for year t = The unemployment rate in year t – the unemployment rate in year $t-1$. (2%)

(b) Using the same Excel spreadsheet, plot the ΔU data on the horizontal axis and $\Delta Y/Y$ data on the vertical axis for 1961-2022 on a scatter plot chart.¹ Now fit a regression line through

¹ See this video (<https://www.youtube.com/watch?v=u7V1LPVIHck>) if you are not sure how to do this.

the data points by adding a linear “trendline” and make the equation of the trendline visible.² Note that this linear trendline is your estimated Okun’s line in Equation (1) for Japan. (2%)

Based on your estimates in part (b), answer the following questions.



(c) What is the estimated value of Okun’s Law coefficient from 1961-2022 for Japan? Type out your answer here (round it to 2 decimal places). Briefly explain what this value means (full credit will not be given for explanation that exceeds 5 lines). (4%).

ANSWER:

The estimated Okun’s Law coefficient = 3.94. Since the slope of the regression line is -3.94 , this implies that a 1 percentage point increase in the unemployment rate is associated with approximately a 3.94 percentage point decrease in real GDP growth. This indicates a negative relationship between unemployment and output growth, consistent with Okun’s Law.

(d) Is the relationship between ΔU and $\Delta Y/Y$ from 1961-2022 for Japan weak or strong? Briefly interpret your answer (full credit will not be given for explanation that exceeds 5 lines). (3%).

ANSWER:

The relationship is negative but relatively weak. The negative slope indicates an inverse relationship between unemployment changes and GDP growth. However, the large spread of observations and the low $R^2(0.089)$ suggest that unemployment changes explain only a small proportion of output fluctuations, implying that other factors also play an important role.

(e) What is the growth rate of potential real GDP from 1961-2022 in Japan? Type out your answer here. Give your response as a percentage rounded to 2 decimal places. Briefly explain what this value means (full credit will not be given for explanation that exceeds 5 lines). (4%).

ANSWER:

The estimated potential GDP growth rate for Japan from 1961–2022 is 3.48%. This is given by the intercept of the Okun’s Law regression and represents the growth rate when the unemployment rate remains unchanged. Deviations from this rate indicate cyclical fluctuations. It reflects the economy’s underlying trend growth and the level of output growth consistent with stable labour market conditions.

² See this video (<https://www.youtube.com/watch?v=u7V1LPVHck>) if you are not sure how to do this.

QUESTION 2 (15%)

The production function of output $Y=AF(K, N)$ for Country Z takes the Cobb-Douglas form and it displays constant returns to scale. Assume $Y=Y_0$, $N=N_0$ and $K=K_0$ initially, and that $A=1$ all the time.

Suppose now there is a change: both labor (N) and capital (K) are increased by 20% to $N=N_1$ and $K=K_1$. Based on the information given, answer the following questions.

(a) How much output will be produced when labor (N) and capital (K) are each increased by 20%? Give your answer by expressing the new output level Y_1 in terms of Y_0 . (2%)

ANSWER:

The Cobb – Douglas production function is

$$Y = AK^\alpha N^{1-\alpha}$$

Given that $A = 1$, the initial output level is:

$$Y_0 = K_0^\alpha N_0^{1-\alpha}$$

After the increase, both capital and labour rise by 20%, so:

$$K_1 = 1.2K_0, N_1 = 1.2N_0$$

The new output level becomes:

$$Y_1 = (1.2K_0)^\alpha (1.2N_0)^{1-\alpha}$$

$$Y_1 = 1.2^\alpha \cdot K_0^\alpha \cdot 1.2^{1-\alpha} \cdot N_0^{1-\alpha}$$

$$Y_1 = 1.2^{\alpha+(1-\alpha)} \cdot (K_0^\alpha N_0^{1-\alpha})$$

$$Y_1 = 1.2^1 \cdot Y_0$$

$$Y_1 = 1.2Y_0$$

Since the production function has constant returns to scale, increasing both labour and capital by 20% increases output by 20%. Therefore, the new output level is $Y_1 = 1.2Y_0$

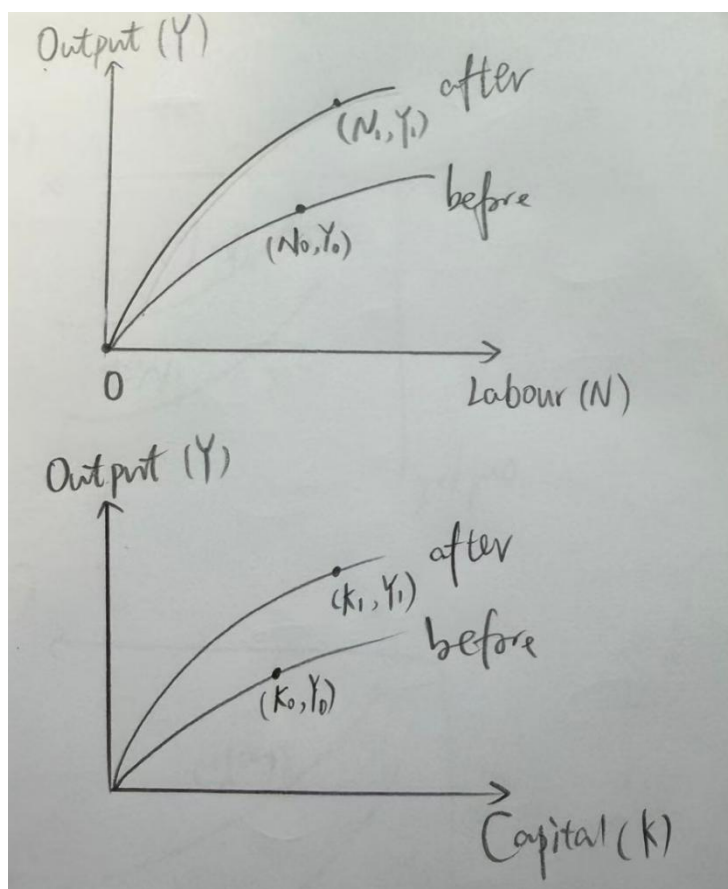
(b) Draw the production function (output versus labor) before the change and the production function (output versus labor) after the change in the same diagram, and indicate on the same diagram the production points (N_0, Y_0) and (N_1, Y_1) , where Y_1 is the level of output produced after the change. The diagram must be well labelled. (4%)

Similarly, draw the production function (output versus capital) before the change and the production function (output versus capital) after the change in the same diagram, and indicate on the same diagram the production points (K_0, Y_0) and (K_1, Y_1) , where Y_1 is the level of output produced after the change. The diagram must be well labelled. (4%)

Briefly explain your diagrams (full credit will not be given for explanation that exceeds 10 lines). (5%).

Note: Screenshot hand-drawn diagrams are fine.

ANSWER:



Because the production function has constant returns to scale, a 20% increase in both labour and capital raises output by 20%, so $Y_1 = 1.2Y_0$.

In the output–labour diagram, the increase in capital shifts the production function upward. At the same time, labour increases from N_0 to $N_1 = 1.2N_0$, so the economy moves to a higher and rightward point on the new curve. In the output–capital diagram, the increase in labour shifts the production function upward. The economy moves from (K_0, Y_0) on the initial curve to (K_1, Y_1) on the new curve, with $K_1 = 1.2K_0$.

In both diagrams, the curves are concave and exhibit diminishing marginal product and shift upward due to the increase in the other input.

QUESTION 3 (12%)

Suppose that recent macroeconomic data reveals the following facts about Canada and Germany:

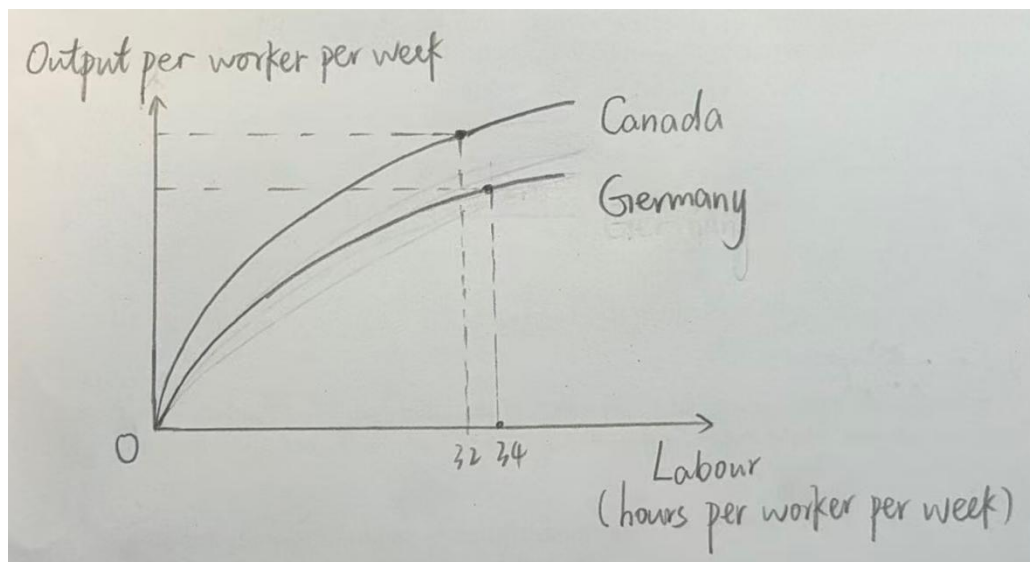
- Canadian labour (measured in hours per worker per week) is more productive than German labour (measured in hours per worker per week);
- German workers are more willing to work than Canadian workers at each real wage rate;
- Germans work longer hours — a German worked an average of 34 hours per week, while a Canadian worked an average of 32 hours per week;
- The real wage rate is higher in Canada than in Germany;
- Canada's output per worker per week is greater than Germany's output per worker per week.

Suppose that the production function takes the Cobb-Douglas form and exhibits diminishing marginal product of labour; labour is measured in hours per worker per week, and the labour market is perfectly competitive. Based on the information given above, answer the following questions.

(a) Draw a well-labelled diagram to show the production function relating output per worker per week and labour (hours per worker per week) for Germany and Canada. Does the German production function lie above the Canadian production function? Explain your answer briefly. Full credit will not be given if your explanation exceeds 5 lines. (Diagram 2% + Explanation 2%)

Note: A screenshot hand-drawn diagram is fine.

ANSWER:

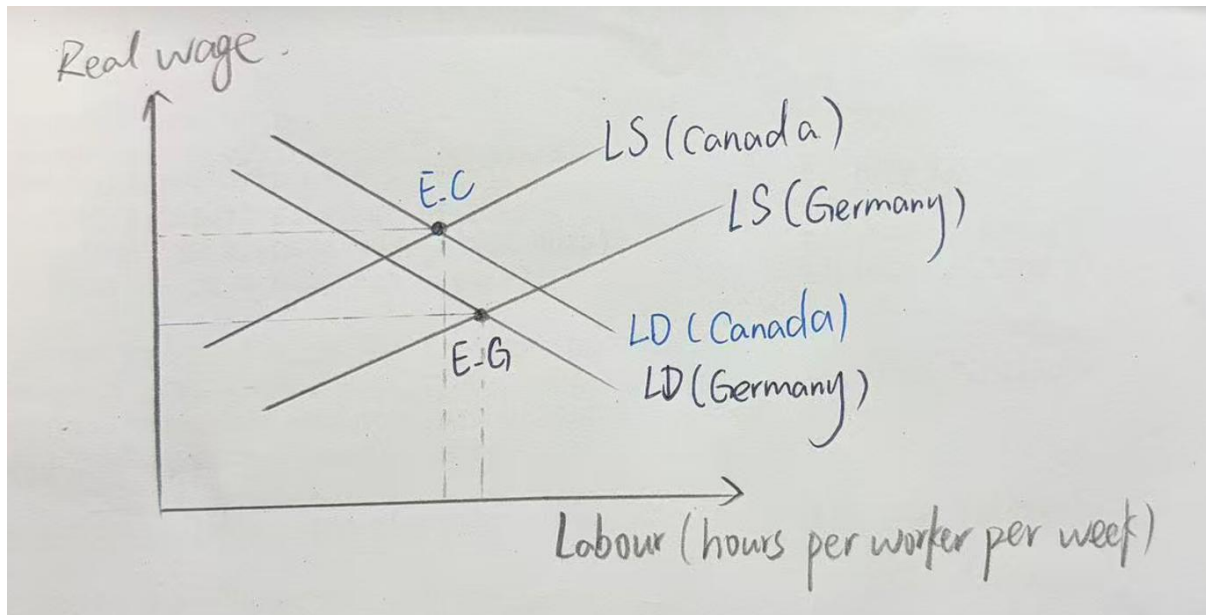


No, the German production function lies below Canada's. Canadian labour is more productive, implying a higher marginal product of labour at each level of hours worked. This may reflect higher technology (A) or capital intensity. Therefore, for any given labour input, Canada produces more output per worker. Germany's longer working hours are due to higher labour supply rather than higher productivity.

(b) Draw a well-labelled labour market diagram to show the labour market equilibrium for Germany and the labour market equilibrium for Canada, where labour is measured in hours per worker per week. Does the German labour demand curve lie to the right of the Canadian labour demand curve? Does the German labour supply curve lie to the right of the Canadian labour supply curve? Explain your answer briefly. Full credit will not be given if your explanation exceeds 10 lines. (Diagram 4% + Explanation 4%)

Note: A screenshot hand-drawn diagram is fine.

ANSWER:



- 1) No, the German labour demand curve does not lie to the right of Canada's. Canadian labour is more productive, so its marginal product of labour is higher at any given employment level. In a competitive market, firms hire until $w = MPL$, therefore, Canada's labour demand curve lies to the right of Germany's.
- 2) Yes, the German labour supply curve lies to the right of Canada's. German workers are more willing to work at each real wage, meaning they supply more hours at any given wage. German workers are more willing to work at each real wage. This results in higher equilibrium hours worked in Germany. In contrast, Canada has a higher equilibrium real wage due to stronger labour demand.

QUESTION 4. (8%)

(a) Use the concept of the income and substitution effects of higher real wage on labor supply to explain the systematic decline in hours worked for a set of countries that have experienced positive growth in per capita income between 1870-2000 in Figure 1. It is reasonable to assume that those countries in Figure 1 have also experienced positive growth in real wage between 1870-2000. Full credit will not be given for explanation that exceeds 10 lines. (5%).

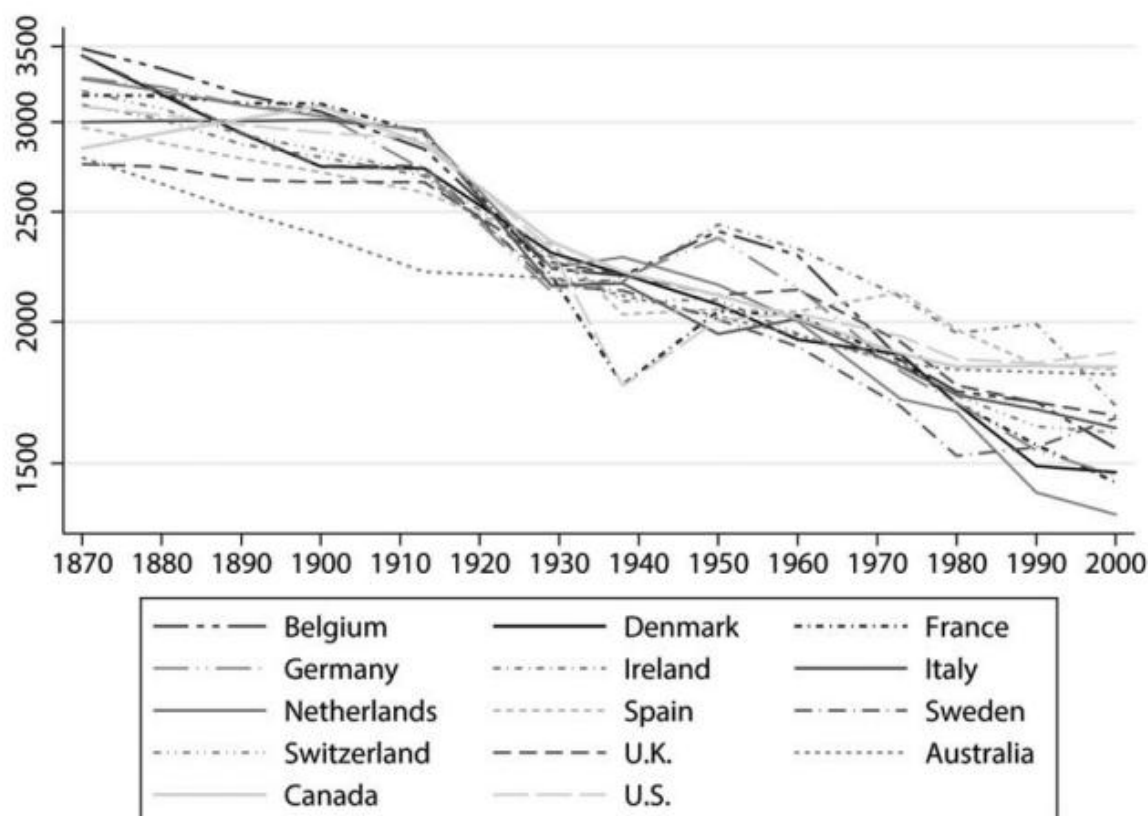


FIG. 1.—Hours worked per worker. The figure shows data for the following countries: Belgium, Denmark, France, Germany, Ireland, Italy, the Netherlands, Spain, Sweden, Switzerland, the United Kingdom, Australia, Canada, and the United States. The scale is logarithmic, which suggests that hours fall at roughly 0.57% per year. Source: Huberman and Minns (2007). Maddison (2001) shows a similar systematic decline in hours per capita. A color version of this figure is available online.

ANSWER:

Figure 1 shows a persistent decline in hours worked despite rising real wages and income. In labour supply theory, a higher real wage generates both substitution and income effects. Individuals choose working time by comparing the marginal benefit of wage income with the marginal cost of forgone leisure. The substitution effect increases labour supply because a rise in the real wage raises the opportunity cost of leisure. However, over the long run, the income effect dominates because leisure is a normal good. As workers become wealthier, they choose more leisure and supply fewer working hours. This leads to a backward-bending labour supply in the long run, explaining the systematic decline in hours worked shown in Figure 1.

(b) Draw a well-labelled diagram with the real wage rate on the vertical axis and hours worked on the horizontal axis (a screenshot hand-drawn diagram is fine) which shows that when real wage is below w_2 , the substitution effect of higher real wage dominates the income effect of higher real wage on labor supply, but when real wage is beyond w_2 , the income effect of higher real wage dominates the substitution effect of higher real wage on labor supply. (3%)

ANSWER:

